

REFLEX with Jev for Efficient Selective Control in LLM Agents

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Abstract

LLM agents often use generative models for bounded decisions, raising the question of when these decisions can be handled more efficiently without reducing task success. We study REFLEX, an agent architecture that uses Jev as a fast, typed decision layer and calls a strong LLM when confidence is low, or generation is required. On a frozen 100-task benchmark, REFLEX achieves 95% success with 72.7% fewer strong-model calls than a strong-only agent, with reductions persisting across three fallback families. Controlled interventions show that reliability depends on action-set size and near-valid alternatives near authorization boundaries. External BFCL and τ -style evaluations reveal limited advantages over a cheap generative cascade when ordinary routing is already highly accurate. These findings identify when selective control with Jev can reduce computation and where its benefits are limited.

1 Introduction

Tool-using language agents increasingly combine reasoning, natural-language generation, and actions through application programming interfaces (APIs). A common design uses a generative large language model (LLM) at nearly every step to inspect the state, select a tool, fill its arguments, decide whether to continue, and produce a response. This design is flexible, but it assigns different kinds of computation to the same model. Many steps require only a choice from a fixed set, suggesting an opportunity to reserve expensive generative reasoning for states that need it.

Existing approaches address related questions at different levels. Multi-LLM routers choose which model should answer an entire query (Chen et al., 2024; Ong et al., 2025; Hu et al., 2024; Li et al., 2026). Tool-use benchmarks primarily evaluate whether a model selects and executes functions correctly (Patil et al., 2025; Wang et al., 2025).

Our question concerns individual decisions within an agent’s trajectory. Can a specialized decision model handle bounded choices while preserving task success, and under what conditions does it offer an advantage over a cheap generative cascade?

We study REFLEX, an agent architecture that uses Jev as its decision layer. Jev is a recently released “System One” model designed to make typed probabilistic decisions rather than generate strings (Almeida, 2026). In REFLEX, Jev returns a bounded choice and a confidence score. Choices with high confidence execute directly, while states with low confidence or requiring free-form generation are passed to the strong LLM. This separates bounded control from generative reasoning within an agent trajectory. Jev may select a tool or continue at the current step, while a strong LLM handles the next step if needed.

On a frozen 100-task controlled benchmark, REFLEX achieves 95% success with 72.7% fewer strong-model calls than a strong-only agent. Across Qwen3.8-Max, Kimi K3, and DeepSeek-V4-Pro fallbacks, it reduces strong-model calls by 66% to 72% while keeping success within a 2-point non-inferiority margin. These results show that selective control with Jev can improve the trade-off between task success and computation across different strong fallback models.

Further experiments clarify which decisions remain difficult. On external BFCL tasks, function selection is already 98.4% accurate, but deciding whether to call any function is only 52.0% accurate. The harder question is therefore often whether the agent has enough authorization or information to act. Controlled interventions show that larger action sets and structurally near-valid alternatives reduce accuracy near authorization boundaries. Changing the type of alternative also shifts errors between irreversible commitment and deferral without a significant change in overall accuracy. These findings show why evaluating only whether

the correct tool was selected can miss important differences in control reliability.

The external multi-turn evaluation also establishes a limit to the efficiency gains. On τ -style tasks, REFLEX reduces cost by a factor of 3.7 relative to a strong-only agent, with a statistically unresolved difference in success. However, a cheap LLM cascade with self-escalation remains competitive. Together with the BFCL results, this suggests that a specialized decision layer has limited room to improve performance when ordinary routing is already highly accurate.

We make four contributions:

- We introduce REFLEX, an agent architecture that uses Jev for bounded decisions and calls a strong LLM when confidence is low, or generation is required.
- We show substantial reductions in strong-model calls on a frozen controlled benchmark across three fallback families, with success within a 2-point non-inferiority margin.
- We identify the conditions that limit the reliability of the decision layer. Larger action sets and near-valid alternatives reduce accuracy on decisions that depend on authorization, while the type of alternative causally shifts errors between unwarranted commitment and deferral.
- We establish the limits of the approach through external BFCL and τ -style benchmark evaluations, showing that a cheap generative cascade remains competitive when ordinary routing is already highly accurate.

These findings show that the value of a specialized decision layer depends on the source of agent difficulty. Using Jev to separate bounded control from generation can reduce computation, but its advantage depends on whether the task contains control decisions that benefit from this separation.

2 Related Work

Cost-aware LLM routing. Cascades and routers use models with different costs and capabilities. FrugalGPT combines prompt adaptation, approximation, and cascades to reduce inference cost (Chen et al., 2024). RouteLLM learns preference-based routers between strong and weak models (Ong et al., 2025). RouterBench and LLMRouterBench provide systematic evaluations of this setting. They show that models can complement one another, but consistently outperforming simple routing baselines remains difficult (Hu et al.,

2024; Li et al., 2026). These methods primarily choose a model for each query. REFLEX separates bounded control from generative reasoning within a single agent trajectory, allowing the two forms of inference to alternate during an episode.

Tool-using agents and evaluation. BFCL evaluates function selection, relevance detection, and, increasingly, multi-turn use of functions by agents (Patil et al., 2025). DialogTool highlights the gap between single-turn tool selection and multi-turn interactions that maintain state (Wang et al., 2025). τ -bench and τ^2 -bench evaluate conversational agents using domain policies and stateful tools. τ^2 adds dual-control environments in which both the user and the agent modify shared state (Yao et al., 2025; Barres et al., 2026). We use these benchmarks to examine specific sources of difficulty. BFCL tests whether ordinary tool selection limits performance, while τ tests whether selective control improves complete trajectories.

Confidence and selective action. Selective prediction uses confidence to balance coverage and risk. Recent work examines confidence in tool-using agents, including model-internal confidence estimators (Subramani et al., 2025) and calibration failures that depend on the tool (Xuan et al., 2026). Other work provides causal evidence that language models use confidence to decide when to abstain (Kumaran et al., 2026). REFLEX uses confidence to guide control. Bounded decisions with high confidence are executed directly, while states with low confidence are delegated to the fallback. We therefore report the relationship between risk and coverage in addition to calibration.

3 REFLEX

3.1 Agent control as a selective decision problem

At agent step t , let s_t denote the observable state and $\mathcal{A}_t$ a typed set of allowed bounded decisions, such as tools or control actions. A decision model J returns

$$(\hat{a}_t, c_t) = J(s_t, \mathcal{A}_t), \quad (1)$$

where $c_t \in [0, 1]$ is the confidence score. REFLEX applies

$$\pi(s_t) = \begin{cases} \hat{a}_t, & c_t \geq \tau \wedge \text{executable}, \\ L_{\text{strong}}(s_t), & \text{otherwise.} \end{cases} \quad (2)$$

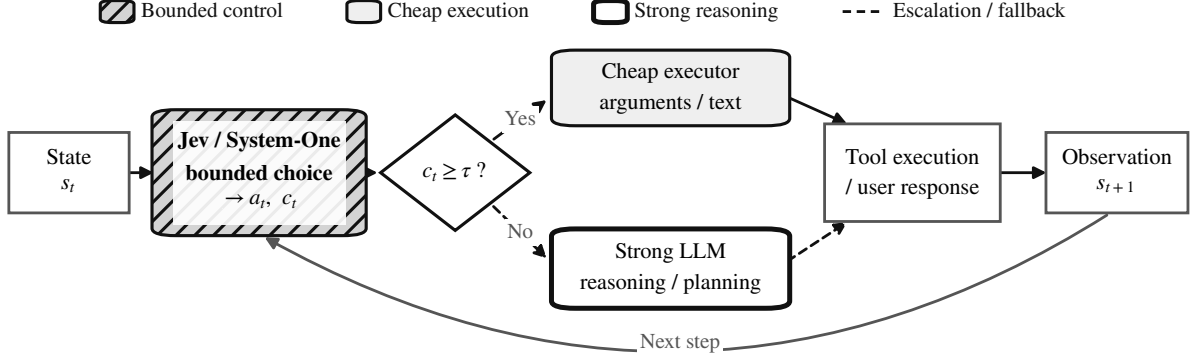


Figure 1: REFLEX separates bounded agent control from open-ended generation. The decision layer reduces strong-model calls at states that can be handled as typed decisions, while retaining the strong LLM as a fallback.

The strong model handles open-ended reasoning, arguments that cannot be obtained deterministically, and user-facing text. In the realistic multi-turn evaluation, we place a cheap generative executor between Jev and the strong model because a typed choice model cannot generate text or arbitrary arguments.

The two outcomes of the gate serve different purposes. A decision with high confidence can remove a strong-model call, while a decision with low confidence passes control to the fallback without forcing an action. REFLEX can therefore retain the fallback’s capabilities while changing which states require it. This motivates the cross-family experiment in § 5. We test whether the decision model identifies a stable subset of states that it can handle autonomously without introducing new failures, rather than whether it can solve every task.

3.2 Decision heads and execution semantics

We distinguish three roles that are often combined in a single agent. *Control* selects among bounded alternatives, such as which tool to call or whether the current state warrants action. *Argument synthesis* supplies structured values for a selected action when those values are not already available in the state. *Generation/reasoning* produces free-form text or resolves open-ended policy questions. REFLEX targets control. In REFLEX-Sim, many arguments are deterministic functions of the observable state. In the external multi-turn setting, a cheap generative executor supplies arguments and text after the control decision, while the strong model remains the final fallback.

This separation guides evaluation. We record tool-selection errors, argument errors, premature termination, and policy errors separately. We also

distinguish *invalid but recoverable* decisions from unsafe actions or actions that cause terminal failure. This allows us to distinguish a harmless extra retrieval from an irreversible write.

Compute metric. We measure the reduction in strong-model calls as

$$\text{GMR} = 1 - \frac{N_{\text{strong}}^{\text{REFLEX}}}{N_{\text{strong}}^{\text{strong-only}}}. \quad (3)$$

We also report cost per episode, cost per successful episode, and strong calls per successful task. These measures help avoid rewarding methods that appear cheap because they fail early.

Selective reliability. For decisions executed autonomously at threshold τ ,

$$\text{Risk}(\tau) = P(\text{invalid} \mid c \geq \tau), \quad (4)$$

$$\text{Coverage}(\tau) = P(c \geq \tau). \quad (5)$$

We summarize this tradeoff using the area under the risk-coverage curve (AURC) and compare it with random escalation at matched coverage.

Confidence gating and self-escalation. A cheap generative model can also decide when to call a stronger model. We include this approach as baseline B3. REFLEX is useful only if its typed decision signal identifies a subset that it can handle autonomously more efficiently or reliably than a cheap LLM cascade. We therefore treat B3 as a central baseline without assuming that a specialized decision layer will outperform it.

4 Experimental Setup

Models. The decision layer uses the version-pinned jev-1.13.0 API. Qwen3.8-Flash and

Qwen3.8-Max serve as the primary small and strong generative models. For cross-family validation, we replace the strong fallback with Kimi K3 or DeepSeek-V4-Pro while keeping Jev, the threshold, tools, and evaluator fixed. Kimi and DeepSeek use the same fixed high reasoning effort in the strong-only and REFLEX conditions.

REFLEX-Sim. We construct a deterministic tool environment that covers direct actions, semantic tool selection, short workflows, policy-governed actions, and ambiguous tasks requiring substantial reasoning (C0 to C4). We score tasks using terminal-state assertions and policy constraints rather than exact matches to a reference trajectory. After evaluator audits, we freeze the final version as REFLEX-SIM v1.2.0. The test set contains 100 held-out tasks, and all reported main runs have zero harness errors.

We compare B0, a strong-only agent, B1, a Flash-only agent, B3, a Flash cascade with self-escalation, and R1, REFLEX with $\tau \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$. The primary operating point is $\tau = 0.5$. Later experiments keep this value fixed without repeating the threshold sweep.

Audit and freeze protocol. The controlled benchmark underwent two evaluator audits before the final v1.2.0 freeze. Success is determined by terminal state and policy constraints rather than agreement with a single gold action sequence, since valid agents may use shorter trajectories than the authored path. All frozen runs retain raw provider payloads, model and version manifests, task hashes, and exact tool traces. Offline grading corrections replay logged actions without regenerating model outputs. The main paper reports only the final frozen scores.

Inference and multiplicity. Unless otherwise stated, confidence intervals use paired cluster bootstrap resampling over the original task or scenario family. All perturbations derived from the same task or family remain within the resampled cluster. Final analyses use 10000 resamples, and exploratory studies use at least 2000. For paired binary episode success, we also use McNemar tests. The RF-5C factorial uses preregistered marginal cardinality, ambiguity, and interaction contrasts, avoiding selection of effects after inspecting the results. Paired uncertainty estimates use task-level bootstrap resampling.

Decision-complexity interventions. To examine the control layer, we vary action-set size K and structural ambiguity. For action a in state s , $d(a | s)$ counts the atomic state or policy facts that must change for a to become correct. We compute this distance from declared conditions that must all hold for an action to be correct. Undeclared actions are classified as FAR. This defines levels of ambiguity independently of model predictions and embedding similarity. RF-5C crosses $K \in \{10, 25, 50\}$ with four ambiguity levels across 60 scenario families and two realizations per cell, yielding 1,440 decisions. We also run a matched intervention with 240 decisions. It replaces a one-change competitor that seeks information with one that makes an irreversible commitment, while holding the gold action, distance, competitor count, K , and semantic similarity fixed.

External evaluations. On BFCL (Patil et al., 2025), we evaluate externally authored function-selection and function-relevance decisions. We expand K using additional functions verified to be distant alternatives. In a τ -style held-out multi-turn validation (Barres et al., 2026), we run 300 episodes comprising 60 paired tasks across five arms with a user simulator. We treat this as external validation rather than a leaderboard submission and use native terminal-state reward as the primary outcome. We audit step-level errors in control selection and argument generation separately.

Pricing and latency accounting. For every API call, we log input and output tokens, model identifiers, decoding parameters, and latency. Costs use dated price snapshots for each provider and region and are recomputed from raw token records. We report agent-side cost for the τ comparison because all arms share the same user simulator. Simulator cost is treated as benchmark overhead rather than a controller-specific difference. REFLEX-Sim also reports strong calls per successful task to account for apparent savings caused by early failure.

5 Results

5.1 REFLEX improves the tradeoff between success and computation

Figure 2 and Table 1 show the corrected frozen results, with costs and additional efficiency measures in Table S14. B0 succeeds on 88% of tasks with 4.10 strong calls per task. R1 at $\tau = 0.5$ achieves 95% success with 1.12 calls per task, a 72.7% re-

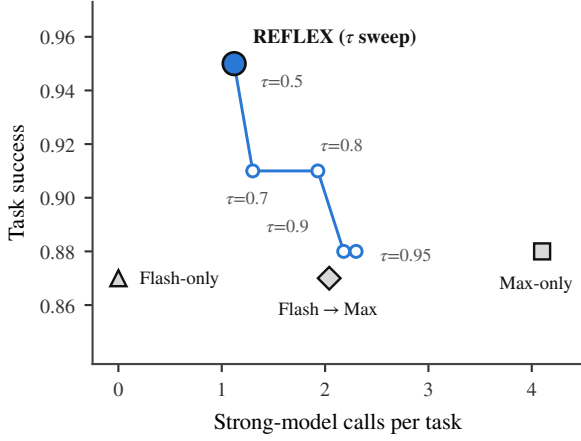


Figure 2: Success and strong-model calls on the frozen 100-task REFLEX-Sim test set. R1 matches or exceeds B0’s success rate at every threshold while using fewer strong-model calls. The largest gain occurs at $\tau = 0.5$.

Condition	Success	Strong calls/task	GMR
B0 Max-only	0.88	4.10	—
B1 Flash-only	0.87	0.00	1.000
B3 Flash→Max	0.87	2.04	0.502
R1 $\tau=0.5$	0.95	1.12	0.727
R1 $\tau=0.7$	0.91	1.30	0.683
R1 $\tau=0.8$	0.91	1.93	0.529
R1 $\tau=0.9$	0.88	2.18	0.468
R1 $\tau=0.95$	0.88	2.30	0.439

Table 1: **Quality–compute frontier on frozen REFLEX-SIM v1.2.0** (100 held-out tasks). GMR is the reduction in strong-model calls relative to B0. Bold marks the preferred frozen operating point ($\tau=0.5$), not a best-in-column.

duction. Higher thresholds reduce the success gain but still use fewer strong calls. At $\tau = 0.9$ and 0.95 , R1 matches B0’s 88% success rate with 2.18 and 2.30 strong calls per task, respectively, corresponding to reductions of 46.8% and 43.9%. The observed success and call-count estimates therefore show a Pareto improvement over B0 at every tested threshold, without higher success at every threshold.

The selective-control metrics in Table S1 show that confidence is useful but imperfect. At $\tau = 0.5$, all 65 episodes without escalation succeed. Of the 251 autonomous decisions on trajectories reached at that threshold, 250 are valid, with one harmless, recoverable exception. The reported AURC is 0.019, compared with 0.145 for random escalation at matched coverage, approximately $7.6\times$ lower using the displayed values. Calibration is moderate, with a Brier score of 0.103 and ECE of 0.129. The main benefit is therefore the ability to distin-

Decision function	n (B0 strong calls)	Replaced by Jev
Tool selection	116	99%
Completion	84	77%
Retrieval decision	160	56%
Clarification	18	28%

Table 2: **Which strong-model decisions the control layer replaces**, on frozen REFLEX-SIM. Thin categories (policy reasoning $n=7$, user-facing generation $n=13$, escalation $n=12$) are excluded from this mechanistic claim; see Appendix for the full breakdown.

Nearest competitor:	None	One at $d=2$	One at $d=1$	Two at $d=1$
$K=10$	0.93	0.92	0.88	0.85
$K=25$	0.87	0.86	0.86	0.82
$K=50$	0.84	0.82	0.83	0.80
	A0	A1	A2	A3
Structural ambiguity level				

Figure 3: RF-5C confirmatory factorial over 60 scenario families. The heatmap shows accuracy by action-set size and structural ambiguity. The cardinality and A0-to-A3 contrasts are significant, while their interaction is not (Table S2).

guish decisions suitable for autonomous execution, despite imperfect calibration.

The paired outcomes in Table S8 help explain the gain. R1 and B0 both succeed on 87 tasks. R1 succeeds on 8 tasks that B0 fails and fails on 1 task that B0 succeeds on, while both fail on 4 tasks (exact McNemar $p = 0.0391$).

Table 2 breaks down the strong-model decisions that are replaced. Replacement rates are much higher for tool selection (99%) and completion (77%) than for clarification (28%), with retrieval between them at 56%. This suggests that routine bounded control is easier to delegate than clarification and judgments that depend on authorization.

5.2 Reliability depends on decision structure

Figure 3 shows the RF-5C factorial, and Table S2 reports the paired contrasts. Averaged over the four ambiguity levels, increasing K from 10 to 50 reduces accuracy by 6.7 percentage points (95% CI $[-11.3, -2.5]$, $p = 0.0008$). Averaged over the three values of K , moving from A0 to A3

reduces accuracy by 5.6 percentage points (95% CI $[-11.9, -0.3]$, $p = 0.038$). The interaction is not statistically significant (+3.3 percentage points, 95% CI $[-3.3, +10.8]$, $p = 0.377$). Each of the 12 cells contains 120 decisions from 60 families with two surface realizations. Contrasts use family-level cell means, with paired cluster bootstrap resampling over families (Table S2). The point estimate for increasing competitor count at fixed distance is larger in magnitude than that for reducing distance at fixed count (-3.3 versus -1.1 percentage points). However, both intervals include zero, with $p = 0.099$ and $p = 0.686$, respectively. These contrasts do not establish that competitor count has a stronger effect than distance.

These effects do not extend to all decision types. Table S9 shows that 192 of the 208 errors occur on control decisions, comprising 117 risk errors and 75 clarification errors. The remaining 16 are completion errors. Retrieval, ordinary tool selection, and workflow decisions have zero errors across their combined 720 decisions. Table S2 reports 21 informative scenario families and 39 at ceiling. In its leave-risk-out sensitivity analysis, the cardinality estimate falls to -2.7 percentage points, with a 95% interval of $[-6.7, +0.5]$ that includes zero. RF-5C, therefore, supports a narrower conclusion. Action-set size and near-valid alternatives matter when the central question is whether the agent has enough authorization or information to commit.

The construction audit in Table S10 also shows why embedding similarity is an inadequate measure of this form of ambiguity. Across 5,931 candidate actions, median similarity to the gold action is 0.310 for one-change competitors, 0.333 for two-change competitors, and 0.237 for far distractors. Moreover, 22.2% of far distractors exceed the median similarity of the pooled near competitors, which is 0.322. These distributions show that similarity under the tested encoder does not consistently follow procedural distance, supporting a definition of ambiguity based on state and policy conditions.

5.3 Competitor type changes the consequences of errors

Figure 4 and Table S3 report the matched intervention. Accuracy is 0.833 with a write competitor and 0.842 with a read competitor, a non-significant difference of $+0.008$ for read minus write (95% CI $[-0.050, +0.067]$, $p = 0.872$). Irreversible-commit errors occur on 10.0% of decisions in the write arm and 1.7% in the read

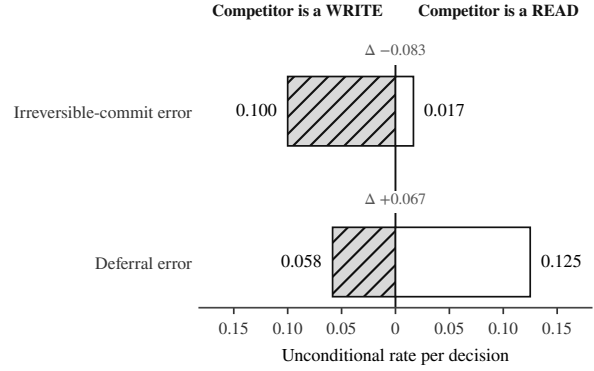


Figure 4: Matched intervention on competitor type with K , gold action, counterfactual distance, competitor count, and semantic similarity held fixed. The intervention changes the *type* of error without a significant change in overall accuracy.

arm ($\Delta = -0.083$, 95% CI $[-0.150, -0.025]$, $p = 0.0014$). Deferral errors occur on 5.8% and 12.5% of decisions, respectively ($\Delta = +0.067$, 95% CI $[+0.025, +0.125]$, $p = 0.0016$). All rates are unconditional per decision, and both differences use read minus write. Competitor type therefore changes the consequences of errors without a statistically resolved change in overall accuracy.

This intervention also revises the exploratory RF-5 observation that deferral was the characteristic failure. In that study, 219 of 366 errors were deferrals (59.8%), and the deferral share rose from 0.36 to 0.72 as K increased. In RF-5C, 24 of 208 errors are identified as deferrals (11.5%, a lower bound), as reported in Table S9. The change across settings motivated retiring the general deferral claim. Table S3 reports no significant asymmetry between the shifts toward commitment and deferral ($\Delta = -0.017$, 95% CI $[-0.067, +0.025]$, $p = 0.534$). This does not establish equal effects or rule out a broader behavioral bias, but it shows that the available competitors influence the observed error pattern. Agents with similar exact-choice accuracy can therefore produce different downstream consequences.

5.4 External benchmarks show where the benefits are limited

Figure 5(a) and Table S4 show a large gap between selecting a tool and deciding whether to act on BFCL. With the same router and function schemas, choosing *which* function to call achieves 98.4% accuracy over 1,800 decisions. Deciding whether *any* function should be called achieves only 52.0% accuracy over 100 relevance cases (Fisher $p =$

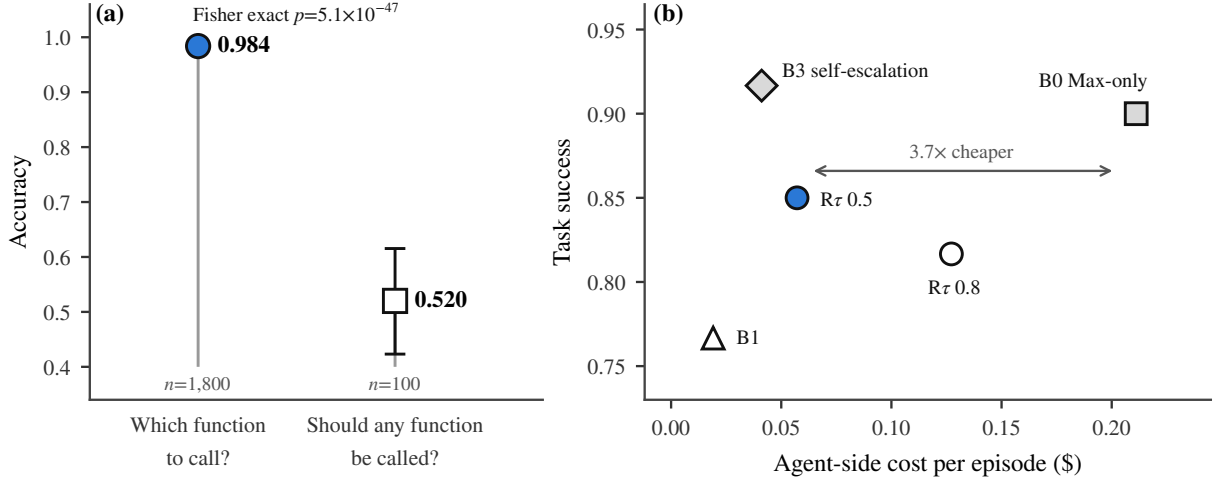


Figure 5: External validation. Ordinary function selection on BFCL is nearly perfect, while deciding whether a function call is relevant remains much harder (left). On τ^2 -bench multi-turn tasks (right), REFLEX costs much less than Max-only, but a simple Flash cascade with self-escalation is competitive.

5.1×10^{-47}). Increasing the candidate set from $K = 2$ to $K = 64$ changes routing accuracy from 99.0% to 98.7%. The paired difference is -0.0033 (95% CI $[-0.010, 0.000]$, $p = 0.7202$), using a paired cluster bootstrap over 300 instances. None of the 29 errors selects an added distant distractor. Incorrect calls nevertheless have a mean confidence of 0.778, indicating that a $\tau = 0.7$ gate would still allow incorrect calls. This external result supports the mechanism observed in RF-5C. The difficult decision is often whether to act, rather than which function to select.

Figure 5(b) and Table 3 show the corresponding system-level comparison on τ^2 -bench v1.0.1 across three domains, with 60 paired tasks evaluated under five conditions. REFLEX at $\tau = 0.5$ reduces per-episode cost by a factor of 3.7 relative to B0 (\$0.0572 versus \$0.2111), with success of 0.850 versus 0.900. Table S5 gives a paired success difference of -0.050 for REFLEX minus B0, with a 95% interval of $[-0.150, +0.050]$ after reversing the reported contrast. B3 is cheaper than REFLEX (\$0.0411) and has numerically higher success (0.917 versus 0.850), but this paired difference is also unresolved ($+0.067$, 95% CI $[-0.033, +0.167]$). Neither comparison establishes equal success. REFLEX at $\tau = 0.8$ is dominated on the observed cost and success estimates by both B3 and REFLEX at $\tau = 0.5$ and is not evaluated further.

The failure audit in Table S11 records 255 successful episodes and 45 failures. Of the failures, 41 are terminal database-state mismatches and four

Condition	Success	\$/episode	\$/success
B0 Max-only	0.900	0.2111	0.235
B1 Flash-only	0.767	0.0191	0.025
B3 self-escalation	0.917	0.0411	0.045
R τ $\tau=0.5$	0.850	0.0572	0.067
R τ $\tau=0.8$	0.817	0.1272	0.156

Table 3: Held-out 300-episode validation (60 paired tasks $\times$ 5 conditions). Only B3–B1 is significant among the reported pairwise contrasts; other tested intervals span zero. Bold marks the best value in each of the first two columns. No control-selection or argument-generation errors occurred in any of the 300 episodes.

are communication failures. There are zero audited control-selection errors and zero argument-generation errors across all 300 episodes. Under these audit categories, the evaluated runs contain no observed routing or argument failures for the specialized decision layer to correct. The result therefore limits claims of a routing advantage on this evaluation, while showing a cost reduction relative to the strong-only baseline.

5.5 Strong-call reductions persist across fallback families

Figure 6 and Tables 4 and S6 report the E6 comparison with Jev and $\tau = 0.5$ fixed while the strong fallback changes. Strong-model calls decrease by 66.1% to 71.9%. The observed success differences for Qwen, Kimi, and DeepSeek are $+0.040$, -0.020 , and 0.000 , respectively. Their 95% bootstrap intervals are $[-0.010, +0.090]$, $[-0.050, 0.000]$, and $[0.000, 0.000]$. No family shows a statistically significant success de-

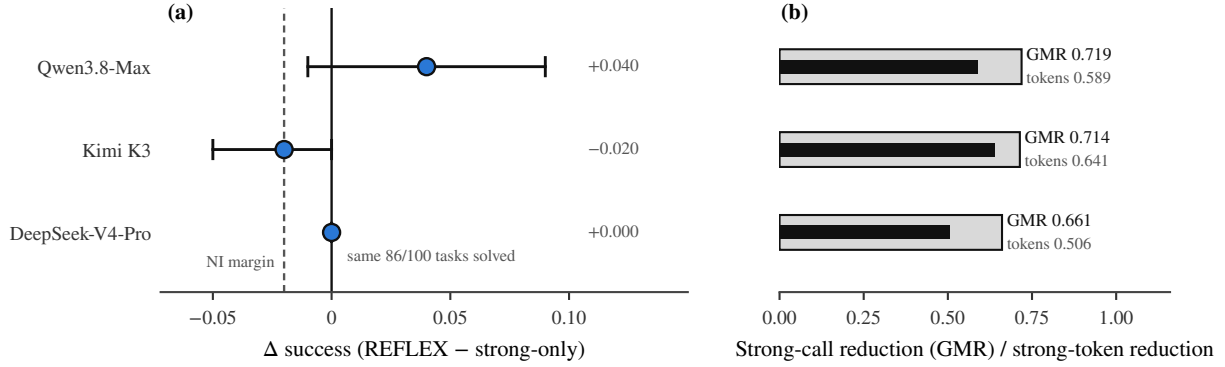


Figure 6: Cross-family validation on the same frozen 100 tasks. Panel (a) shows paired success differences and 95% bootstrap intervals. The dashed line marks the -0.02 margin, which Kimi’s interval crosses. Panel (b) shows strong-call and strong-token reductions. Strong-model calls decrease by 66% to 72% across the three fallback families.

crease. However, the preregistered acceptance rule checked only whether the point estimate satisfied $\Delta \geq -0.02$. It was not a formal non-inferiority test. Under the confidence-bound rule requiring the lower 95% bound to exceed -0.02 , Qwen and DeepSeek meet the criterion, while Kimi does not (Table S12). Kimi’s strong-only arm achieves 100% success and REFLEX achieves 98%, so its call reduction accompanies an observed two-task loss.

Figure 6(b) and Table S6 also show strong-token reductions of 50.6% to 64.1%, indicating that call savings are accompanied by substantial token savings. E6 and the main threshold sweep are separate executions, and each is compared with its own baseline. Table S13 documents their differences. Qwen strong-only success is 0.88 in the main run and 0.91 in E6, with six tasks lost and nine gained. Its strong-call rate changes from 4.10 to 3.95. REFLEX succeeds on 95 tasks in both runs, but four tasks are lost and four gained, while the autonomous subset changes from 65 to 66 tasks. The configurations differ only in the price table and Jev retry settings, neither of which affects B0. The B0 variation therefore occurs despite the same tasks, prompts, and temperature-zero inference condition. This provider variation limits the interpretation of small differences between separate runs.

Table 5 shows a task-level pattern that aggregate success does not reveal. For every fallback family, the same 66 of 100 tasks are handled entirely by Jev, and all 66 succeed. Every observed REFLEX failure in E6 occurs within the 34-task subset requiring escalation. For DeepSeek, strong-only and REFLEX succeed on exactly the same 86 tasks. REFLEX thus preserves the set of tasks solved by

the fallback while removing two-thirds of its calls. The trace audit in Table S12 further examines the failures and decisions within escalated episodes.

5.6 What does the gate identify?

The stable subset in Table 5 provides information beyond aggregate success differences. The gate does not adapt to the identity or capability of the downstream strong model. With the decision layer and threshold fixed, the same autonomous subset is observed in all three E6 runs, and all 66 tasks succeed in each run. These rows describe one shared task subset, not three independent samples. This E6 subset is distinct from the 65 no-escalation episodes in the main threshold run reported in Table S1. Differences in model capability appear on the remaining 34 tasks. Under strong-only execution, Kimi solves all of them, Qwen solves 76.5%, and DeepSeek solves 58.8%. REFLEX, therefore, filters computation according to the task rather than learning to approximate a particular fallback.

GMR and autonomous task coverage use different denominators, so their numerical difference alone does not locate the savings. The trace analysis in Table S12 provides direct evidence. Within episodes that escalate at least once, Jev still handles 26.2% of steps for Qwen, 26.8% for Kimi, and 21.3% for DeepSeek without a strong-model call. The savings therefore extend beyond the 66 fully autonomous tasks. Across the three REFLEX arms, the audit identifies 21 failures, comprising five for Qwen, two for Kimi, and 14 for DeepSeek. None contains an incorrect autonomous decision. The audit attributes these observed failures to fallback handling after escalation rather than an incorrect decision executed by Jev.

Fallback family	Strong-only	REFLEX	Δ	GMR
Qwen3.8-Max	0.91	0.95	+0.04	0.719
Kimi K3	1.00	0.98	−0.02	0.714
DeepSeek-V4-Pro	0.86	0.86	+0.00	0.661

Table 4: Cross-family robustness. Within each family the identical strong-model configuration (including reasoning effort) is used for both the strong-only baseline and REFLEX. All individual success differences are non-significant. The predeclared criterion compared the *point estimate* of Δ to a -0.02 margin; a formal non-inferiority test, which requires the lower confidence bound to clear that margin, passes for Qwen and DeepSeek but not for Kimi (Table S12). Kimi strong-only is at ceiling, so that row cannot test quality improvement at all. Baseline success here is not comparable to Table 1: both are the same condition executed twice (Table S13).

Fallback	Autonomous n	Autonomous success	Escalated n	Strong-only on escalated
Qwen3.8-Max	66	1.000	34	0.765
Kimi K3	66	1.000	34	1.000
DeepSeek-V4-Pro	66	1.000	34	0.588

Table 5: The fixed gate selects the same 66-task autonomous subset regardless of fallback family, and all observed E6 failures occur in the escalated subset. The three rows describe the same task subset under one frozen gate, not three independent samples.

6 Discussion

The experiments show that the value of specialized decision layers depends on the setting. First, bounded control can serve as a separate form of computation. On REFLEX-Sim, a stable 66% subset is solved autonomously with zero failures across three distinct fallback families, while the strong model handles the remaining states. This substantially reduces computation without requiring the decision model to reproduce the fallback’s full reasoning behavior.

The cross-family results show large call reductions without a statistically significant success decrease, but this does not establish non-inferiority for every family. Qwen and DeepSeek meet the lower-confidence-bound criterion, while Kimi does not. The repeated Qwen runs also show a three-percentage-point change in strong-only success despite the same inference condition (Table S13). This observed change is comparable in scale to the E6 success differences. It is evidence of run-to-run variability, not an estimated uncertainty band, and limits claims about small quality differences from a single execution.

Second, the advantage depends on the source of difficulty. BFCL shows that the tested decision model already selects the correct function almost perfectly, while deciding whether a call is warranted remains harder. The τ^2 -bench validation extends this finding to complete trajectories. When no tool-selection or argument errors are observed, a dedicated router has little room to improve per-

formance, and a simple, cheap cascade with self-escalation is competitive. This agrees with broader LLM-routing evidence that more complex routers do not consistently outperform simple baselines under a common evaluation (Li et al., 2026).

Third, accuracy alone can hide differences in control errors. Near authorization boundaries, larger candidate sets and greater structural ambiguity reduce reliability. Changing the *type* of one near-valid competitor also shifts errors between irreversible commitment and deferral without a significant change in accuracy. Evaluation of selective agents should therefore report action consequences and the types of risk, as well as exact-choice accuracy.

These results suggest three conditions for using specialized decision layers. A meaningful part of the trajectory should consist of bounded, repeated decisions. Confidence should distinguish states suitable for autonomous execution from those that require escalation. The downstream benchmark should also account for uncertainty in control decisions. Otherwise, the additional component may increase complexity without improving the cost-success trade-off.

The full claim register in Table S15 records the supported and retired claims, including the qualified cross-family and τ^2 -bench comparisons.

Why the negative external result matters. REFLEX-Sim alone would suggest broad benefits from the architecture. BFCL and τ^2 -bench show that the room for improvement depends on the task.

This matters for both deployment and evaluation. A benchmark with almost no routing errors cannot establish the value of a new routing layer, and overall agent success does not reveal whether failures arise from control, arguments, or downstream reasoning. The negative external result narrows the claim and makes its limits clearer and more directly testable.

Implications for agent benchmarks. Across our experiments, ordinary tool selection often reaches ceiling performance, while decisions that depend on authorization remain difficult. Future agent benchmarks should therefore annotate both the correct action and why it is currently valid. Relevant information includes the required evidence, authorization state, reversibility, and whether seeking more information is still warranted. These annotations would support the evaluation of selective risk without requiring one reference trajectory.

7 Conclusion

We presented REFLEX, which separates bounded control from open-ended reasoning in LLM agents. In the cross-family comparison, it reduces strong-model calls by roughly two-thirds across Qwen, Kimi, and DeepSeek fallbacks, with observed success differences ranging from a two-percentage-point loss to a four-percentage-point gain. No success decrease is statistically significant, but only Qwen and DeepSeek meet the confidence-bound non-inferiority criterion in these runs. Controlled interventions show that reliability is mainly limited by commitment decisions that depend on authorization, rather than ordinary tool selection. BFCL and τ^2 -bench provide a complementary negative result. Cheap-model cascades are competitive when routing is already nearly perfect. The value of a decision layer, therefore, depends on whether its form of inference fits the computation needed at each step.

Limitations

We use one proprietary typed decision model, Jev, so the results do not establish that all non-generative decision models have the same selective-control behavior. E6 varies the strong fallback across three families but keeps the decision model fixed. Several controlled benchmarks also show ceiling effects for 2026 frontier models. Kimi reaches 100% success on REFLEX-Sim, and ordinary BFCL function selection is nearly perfect.

The effects of action-set size and ambiguity in RF-5C are concentrated in authority and risk families and should not be generalized to all agent decisions. The τ evaluation is a held-out validation with 300 episodes, rather than a full leaderboard evaluation with multiple trials. Its native reward can also miss some procedural differences. We therefore use it as external evidence about cost and where failures occur, without claiming benchmark superiority. Finally, reported API costs are dated snapshots and may change with provider pricing, caching, or deployment region.

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A Hierarchical Routing Ablation

We tested a hierarchical routing variant that first selects a tool family and then selects a tool within that family. This design reduces the number of alternatives considered at the second stage, but introduces a separate family-selection decision. The ablation examines whether this additional stage helps when the candidate set is large.

An initial smoke study suggested that hierarchical routing performed worse than flat routing. However, the implementation audit identified differences in the information shown to the compared systems (Table S7). The family-selection stage received bare family names without their authored descriptions, while the router described only the original pilot tools rather than every candidate. In addition, a generative model prompt silently omitted candidates without a known specification. These issues weakened the hierarchical arm and affected the interpretation of the flat-routing comparison. The affected smoke results, therefore, do not support a clean comparison between routing structures.

The repair supplied the authored family descriptions, described all candidates, and retained candidates by identifier when a specification was unavailable. After these repairs, hierarchical accuracy improved by 5 to 10 percentage points relative to the broken implementation. This improvement measures the effect of repairing the implementation and does not establish an advantage over corrected flat routing. Conditional on selecting the correct family, within-family tool selection was perfect in the evaluated probes. The remaining hierarchical errors were concentrated at family boundaries, indicating that the first stage remained a source of failure.

Corrected flat routing was already near the ceiling on well-posed single-step probes, including those with large candidate sets. Splitting a decision into two stages, therefore, offered limited room for improvement while introducing a potentially ambiguous family assignment. An action may plausibly fit more than one semantic category, so reducing the number of tools considered does not necessarily make the full decision easier. We consequently treat hierarchy as a negative ablation rather than part of REFLEX. The result motivated RF-5C’s controlled study of action-set size and structural ambiguity instead of a larger hierarchy sweep. The general claim that hierarchical routing helps at large candidate-set sizes is retired in Table S15.

B Additional Evaluation Results

B.1 Selective reliability and confidence

Table S1 reports operational metrics from each executed threshold run. These measurements describe the trajectories reached under that threshold, rather than counterfactual decisions pooled across runs. Decision risk is the fraction of autonomous decisions that are invalid. Episode-level MER instead measures the fraction of episodes containing at least one invalid autonomous decision. A recoverable invalid decision can therefore increase both measures without causing the episode to fail.

At $\tau = 0.5$, 250 of 251 autonomous decisions are valid, giving a decision risk of 0.0040. The single invalid decision is recoverable, and all 65 episodes completed without escalation succeed. Increasing the threshold reduces decision coverage from 0.691 at $\tau = 0.5$ to 0.439 at $\tau = 0.95$. No unsafe autonomous decision is observed at any reported threshold. These are empirical outcomes

on the evaluated runs, not guarantees of error-free autonomous execution.

Jev’s AURC is 0.019, compared with 0.145 for random escalation at matched coverage. Its Brier score of 0.103 and ECE of 0.129 show that useful selective discrimination does not require perfect calibration. B3 does not emit a numeric confidence score, so an equivalent confidence-based risk-coverage curve is unavailable for that baseline.

B.2 RF-5C design and statistical contrasts

RF-5C crosses three action-set sizes, $\mathcal{K} = \{10, 25, 50\}$, with four ambiguity levels, A0 to A3. Each of the 60 scenario families contributes two surface realizations to each cell, yielding 120 decisions per cell and 1,440 decisions overall. The ambiguity levels contain no near competitor, one two-change competitor, one one-change competitor, and two one-change competitors, respectively.

Let $m_f(K, A)$ be the mean correctness of the two surface realizations for family f in cell (K, A) , and let $\mathcal{L} = \{A0, A1, A2, A3\}$. With $F = 60$, the reported cardinality and ambiguity contrasts are

$$\Delta_K = \frac{1}{4F} \sum_{f=1}^F \sum_{A \in \mathcal{L}} [m_f(50, A) - m_f(10, A)], \quad (6)$$

$$\Delta_A = \frac{1}{3F} \sum_{f=1}^F \sum_{K \in \mathcal{K}} [m_f(K, A3) - m_f(K, A0)]. \quad (7)$$

Thus, the ambiguity contrast averages over all three action-set sizes rather than conditioning on $K = 25$. The cardinality contrast averages over all four ambiguity levels. Confidence intervals use 10,000 paired cluster bootstrap resamples over scenario families with seed 20260921. Each resampled family retains its complete set of cells. The fixed contrast weights are recorded in `rf5c/inference_plan.json`.

Table S2 reports significant marginal decreases for cardinality and ambiguity, but not their interaction. The count-at-fixed-distance and distance-at-fixed-count estimates are also individually non-significant. Their different point estimates, therefore, do not establish which component has the larger effect. The leave-risk-out result is a sensitivity analysis rather than a preregistered primary contrast. Its interval includes zero, showing that the aggregate cardinality result depends on the decision families represented in the panel.

B.3 Matched intervention on competitor type

The matched intervention contains 240 decisions at $K = 25$. It changes whether the single one-change competitor is an information-seeking read or an irreversible write, while holding the gold action, competitor count, procedural distance, and semantic similarity fixed. Table S3 reports every difference as the read-arm rate minus the write-arm rate. Error rates are unconditional per decision, not conditional on making an error.

Accuracy is 0.833 in the write arm and 0.842 in the read arm. The difference is unresolved, with a 95% interval of $[-0.050, +0.067]$. However, the read arm has fewer irreversible-commit errors and more deferral errors. The corresponding differences are -0.083 and $+0.067$, and both intervals exclude zero. Competitor type, therefore, changes the consequences of errors even when the overall accuracy difference is not statistically resolved. The asymmetry test has $p = 0.534$, which does not establish that the two shifts are equal or that the model has no broader behavioral bias.

B.4 BFCL function selection and relevance

Table S4 separates function selection from the decision of whether a function applies. The cardinality evaluation contains 300 instances at each of six candidate-set sizes, for 1,800 decisions in total. Added functions are certified as topically distant and checked through a manual sample audit. The resulting comparison tests expansion with distant alternatives and do not establish robustness to arbitrary near-valid competitors.

The paired accuracy difference between $K = 64$ and $K = 2$ is -0.0033 , with a 95% interval of $[-0.010, 0.000]$ and $p = 0.7202$. Resampling is paired and clustered over the 300 instances. None of the 29 incorrect selections chooses an injected distant distractor. In contrast, accuracy on the separate 100-case relevance evaluation is 0.520, compared with 0.984 for function selection. Incorrect calls have a mean confidence of 0.778, indicating that a threshold of 0.7 cannot exclude all incorrect calls. The construction uses procedural checks because high embedding similarity alone cannot establish whether a function is a valid alternative.

B.5 Paired comparisons on τ^2 -bench

The external multi-turn evaluation uses τ^2 -bench v1.0.1 across three domains. Sixty held-out tasks are evaluated under five conditions, yielding 300

τ	Coverage	Decision risk	UnsafeRisk	NES	RecoveryRate	Episode-level MER
0.5	0.691	0.0040	0.0000	1.000	0.857	0.010
0.7	0.629	0.0045	0.0000	1.000	0.800	0.010
0.8	0.514	0.0049	0.0000	1.000	0.827	0.010
0.9	0.468	0.0000	0.0000	1.000	0.769	0.000
0.95	0.439	0.0000	0.0000	1.000	0.769	0.000

Table S1: **Operational selective-control metrics**, each computed from its own executed threshold run (no pooling of counterfactual replays). Decision risk is the invalid rate among autonomous decisions; episode-level MER is the fraction of episodes containing at least one invalid autonomous decision. The two levels are reported separately and are not the same statistic: at $\tau=0.5$, 250 of 251 autonomous decisions were valid (99.6%), and the single invalid decision was recoverable, while 65/65 no-escalation episodes succeeded (100%). Decision-level validity is not perfect; episode-level no-escalation success is. Jev AURC = 0.019; random matched-coverage AURC = 0.145; Brier = 0.103; ECE = 0.129. A comparable curve for the B3 router is unavailable: it emits no numeric confidence.

Contrast	Estimate	95% CI	p
K 10 $\rightarrow$ 50 at mean ambiguity	-0.067	[-0.113, -0.025]	0.0008
A0 $\rightarrow$ A3 at $K=25$	-0.056	[-0.119, -0.003]	0.038
$K \times A$ interaction	+0.033	[-0.033, +0.108]	0.377
distance at fixed count	-0.011	[-0.061, +0.031]	0.686
count at fixed distance	-0.033	[-0.081, +0.003]	0.099
leave-risk-out (K effect)	-0.027	[-0.067, +0.005]	–

Table S2: **RF-5C factorial statistics**. 60 scenario families, of which 21 are informative and 39 sit at ceiling. 1,440 decisions in 12 cells of 120 (60 families $\times$ 2 surface realizations). The estimator is a paired cluster bootstrap over scenario family, 10,000 resamples, seed 20260921: the two surface realizations are first averaged into each family’s cell mean, then each contrast is a fixed weighted sum of those cell means, with weights given in rf5c/inference_plan.json. The K contrast is the mean over the four ambiguity levels of ($K=50 - K=10$), i.e. $0.825 - 0.892 = -0.067$ from the cell means in Figure 3; the ambiguity contrast is the mean over the three K levels of ($A3 - A0$). The leave-risk-out row shows the aggregate effect is carried by control/authority-sensitive families and must not be generalized across all decision types (Table S9); Its p is not reported because that refit is a sensitivity analysis, not a pre-registered contrast.

episodes. Table S5 uses paired cluster bootstrap resampling over tasks with 10,000 resamples. Pairing preserves the comparison between conditions on the same task.

Among the reported contrasts, only B3 minus B1 has an interval excluding zero. B3 exceeds REFLEX at $\tau = 0.5$ by an observed 0.067, but the interval is $[-0.033, +0.167]$. B0 exceeds REFLEX by 0.050, with an interval of $[-0.050, +0.150]$. These unresolved differences do not establish equal performance. REFLEX’s lower episode cost relative to B0 can therefore be reported separately from the uncertainty in the difference in their success. The failure categories underlying these outcomes are given in Table S11.

B.6 Cross-family calls, tokens, and costs

Table S6 reports the E6 results for each strong fallback. Within a family, the strong-only and REFLEX arms use the same strong-model configuration, including reasoning effort. Jev and the threshold remain fixed across families. Strong-call reduc-

tions are 71.9% for Qwen, 71.4% for Kimi, and 66.1% for DeepSeek. Strong-token reductions are 58.9%, 64.1%, and 50.6%, respectively.

Call reduction, token reduction, and monetary cost reduction are different quantities. A removed call can vary in length, and the remaining calls can have a different token distribution. Verified cost reduction is reported only for Qwen at 52.2%. Missing Kimi and DeepSeek cost entries indicate that a verified frozen provider price is unavailable, not that cost reduction is zero. The success intervals and their implications for non-inferiority are examined further in Table S12.

C Implementation Audits and Verification

C.1 Corrections before the final analyses

Table S7 records the issues found during development, their potential effects, and the evidence affected. The pilot evaluator initially required an exact ticket-reason string, which could penalize

Outcome	Commit-shaped	Defer-shaped	Δ (defer–commit)	95% CI	p
Accuracy	0.833	0.842	+0.008	[−0.050, +0.067]	0.872
Irreversible-commit error	0.100	0.017	−0.083	[−0.150, −0.025]	0.0014
Deferral error	0.058	0.125	+0.067	[+0.025, +0.125]	0.0016
Shape-pull asymmetry	–	–	−0.017	[−0.067, +0.025]	0.534

Table S3: **Matched competitor-shape intervention**, 240 decisions at $K=25$. Arms differ only in whether the single one-change competitor is a read or a write; all rates are unconditional per decision.

Stratum	Accuracy	n
<i>Routing cardinality</i>		
$K=2$	0.990	300
$K=4$	0.990	300
$K=8$	0.980	300
$K=16$	0.973	300
$K=32$	0.983	300
$K=64$	0.987	300
<i>Relevance</i>		
Which function to call	0.984	1,800
Should any function be called	0.520	100
False-positive abstention	0.010	100
Mean confidence of wrong calls	0.778	–
<i>Distractor audit</i>		
Errors selecting an injected distant distractor	0 / 29	–

Table S4: **BFCL**. Controlled fill was certified topically distant and hand-audited on a random sample. No plausible-distractor regime is reported: BFCL’s native distractors reach 0.94 similarity to their gold, so similarity alone cannot certify that an injected function is irrelevant.

otherwise valid outcomes. Other pilot issues included unobservable eligibility conditions, omitted action histories, and argument templates that covered only the authored path. These issues were corrected through evaluator changes, task repairs, and more complete agent inputs. Offline grading corrections used recorded trajectories rather than regenerating model outputs.

The table also records the candidate-description and prompt-display errors relevant to the hierarchical ablation. Later checks addressed cache keys that omitted decoding parameters, Jev tokens priced at strong-model rates, and tie handling in the BFCL AUROC calculation. Entries marked as affecting no frozen evidence were found before the corresponding module was frozen. The floating-point tolerance repair applies to a point-estimate acceptance check and does not turn that check into a statistical non-inferiority test.

C.2 Paired outcomes on REFLEX-Sim

Table S8 gives the full paired outcome table for B0 and REFLEX at $\tau = 0.5$ on the 100-task main test set. Both succeed on 87 tasks. REFLEX succeeds on eight tasks that B0 fails and fails on one task that B0 succeeds on. Both fail on four tasks.

The margins are therefore 88 B0 successes and 95 REFLEX successes. The exact two-sided McNemar test uses the nine discordant pairs and gives $p = 0.0391$. This pairing supports a net improvement in the observed main run while retaining the one observed regression.

C.3 RF-5C error concentration and exploratory deferrals

Table S9 separates the 1,440 RF-5C decisions into six types, each containing 240 decisions. Risk contributes 117 errors and clarification contributes 75, so control decisions account for 192 of the 208 errors. The remaining 16 are completion errors. Retrieval, ordinary tool selection, and workflow have no errors across their combined 720 decisions. The full analysis still includes all 60 scenario families, although 39 remain at ceiling and 21 are informative for the observed errors.

The exploratory RF-5 study recorded 219 deferrals among 366 errors (59.8%), with the share increasing from 0.36 to 0.72 as K increased. RF-5C identifies 24 deferrals among 208 errors (11.5%), reported as a lower bound. These are different study settings and do not form a matched estimate of a change in deferral rate. Their disagreement, to-

Contrast	Δ success	95% CI	Significant?
B3 – B1	+0.150	[+0.033, +0.267]	yes
B3 – R τ 0.5	+0.067	[−0.033, +0.167]	no
B0 – R τ 0.5	+0.050	[−0.050, +0.150]	no
R τ 0.5 – R τ 0.8	+0.033	[−0.067, +0.133]	no
B3 – B0	+0.017	[−0.083, +0.117]	no
Audit: $E_{\text{control}} = 0$, $E_{\text{arguments}} = 0$ across all 300 episodes.			

Table S5: Paired cluster bootstrap over the 60 held-out tasks, 10,000 resamples. Only B3–B1 excludes zero.

Strong family	Strong-only	REFLEX	Δ	95% CI	B0 calls/task	REFLEX calls/task	GMR	Token red.	Cost red.
Qwen3.8-Max	0.91	0.95	+0.04	[−0.01, +0.09]	3.95	1.11	0.719	0.589	0.522
Kimi K3	1.00	0.98	−0.02	[−0.05, +0.00]	3.53	1.01	0.714	0.641	–
DeepSeek-V4-Pro	0.86	0.86	+0.00	[+0.00, +0.00]	4.04	1.37	0.661	0.506	–

Table S6: Full E6 results. Cost reduction is reported only where a verified frozen provider price exists. *Kimi strong-only is at ceiling (1.00), so that row cannot test quality improvement.* The predeclared non-inferiority criterion was applied to the point estimate of Δ , not to the confidence bound; Table S12 gives the formal test.

gether with the competitor-type intervention, limits the claim that deferral is a general failure tendency of the router.

C.4 Procedural distance and embedding similarity

Table S10 examines all 5,931 candidate actions from the 60 RF-5C families using cosine similarity under all-MiniLM-L6-v2. There are 180 one-change competitors, 120 two-change competitors, and 5,631 far distractors. Their median similarities to the gold action are 0.310, 0.333, and 0.237, respectively. Thus, the median for two-change competitors is higher than that for one-change competitors, even though the procedural distance between them is greater.

The distributions also overlap. Of the far distractors, 22.2% exceed the pooled near-competitor median similarity of 0.322. Similarity under this encoder, therefore, does not consistently order procedural validity. The result supports the use of declared state and policy conditions to construct ambiguity levels, while remaining specific to the evaluated candidates and encoder.

C.5 Failure categories on τ^2 -bench

Table S11 accounts for every episode in the external evaluation. Of 300 episodes, 255 succeed and 45 fail. Forty-one failures are terminal database-state mismatches, and four are communication failures. The latter occur once in B3, once in REFLEX at $\tau = 0.5$, and twice in REFLEX at $\tau = 0.8$.

The audit records zero control-selection errors

and zero argument-generation errors in every condition. This distinction matters because a failed episode is not necessarily a failed routing decision. The evidence shows no observed control or argument failures under the applied audit categories. It does not establish that routing can never influence success in other tasks or deployments. The reported comparison, therefore, concerns the cost and observed failure structure of this evaluation rather than a general routing advantage.

C.6 Non-inferiority and decisions within escalated episodes

Table S12 separates the original acceptance rule from a confidence-bound assessment. The preregistered rule compared the observed difference in success with -0.02 . All three families meet this point-estimate rule. The confidence-bound criterion instead requires the lower 95% bound to be strictly greater than -0.02 . Qwen meets it with a lower bound of -0.010 , and DeepSeek meets it with a lower bound of 0.000 . Kimi’s lower bound is -0.050 , so its non-inferiority is not established by this criterion. The absence of a statistically significant decrease is therefore not interchangeable with evidence of non-inferiority.

The same table measures Jev’s contribution within episodes that escalate at least once. Jev handles 26.2% of steps in Qwen episodes, 26.8% in Kimi episodes, and 21.3% in DeepSeek episodes without reaching the strong model. These step-level measurements directly show that savings extend beyond the 66 fully autonomous tasks. They

Stage	Issue found	Potential bias	Resolution	Frozen evidence affected
Pilot grading	Exact-string <code>create_ticket</code> reason match	Inflated recovery counts	Evaluator relaxed; offline re-grade	regraded to v1.2.0
Pilot task design	C3 eligibility unobservable from state	Ill-posed decision	Observable fields added; tasks rebuilt	pre-v1.2.0 only
RF-5a	E1a: only pilot tools described to the router	Weakened added distractors; biased K sweep flat	All candidates described; parity logged per call	RF-5a smoke only
RF-5a	E1b: family stage sent bare family names	Understated hierarchical arm	Authored descriptions wired in	RF-5a smoke only
RF-5a	E1c: LLM prompt silently dropped unknown candidates	Model graded on options it never saw	Never drop; list by id if unspecified	RF-5a Flash arm void
Pilot agents	Prompt omitted action history	Repeated-action loops	History added	pre-v1.2.0 only
Pilot agents	Argument templates covered gold-path tools only	Non-gold tools always failed	Templates extended; LLMs propose args	pre-v1.2.0 only
E6	Provider cache key omitted decoding parameters	A reply from one reasoning effort could serve another	Decode params and extra_body folded into the key	none
E5/E6	Jev tokens priced at the strong-model rate	Overstated REFLEX cost	Tokens split by controller	none
BFCL	AUROC ignored ties	Sign-flip asymmetry in a tied score	Mid-ranks	none
E6	Non-inferiority check used a literal ≥ -0.02	Criterion reported false on a float artefact	Tolerance added	none

Table S7: **Audit history.** Issues that materially affected interpretation, and which frozen evidence each one touches. Items marked “none” were found before the affected module was frozen.

B_0 Max-only	REFLEX $\tau=0.5$		
	succeeds	fails	total
succeeds	87	1	88
fails	8	4	12
total	95	5	100

Table S8: **Paired task outcomes** on the frozen 100-task REFLEX-SIM test split. The gate recovers 8 tasks B_0 fails and loses 1, so the marginal move from 0.88 to 0.95 is not a trade of one failure set for another. McNemar exact $p=0.039$ on the 9 discordant pairs.

use a different denominator from autonomous task coverage and from GMR.

Across the REFLEX arms, five Qwen episodes, two Kimi episodes, and 14 DeepSeek episodes fail. None of these 21 episodes contains an incorrect autonomous decision. The trace audit attributes the observed failures to fallback handling after escalation. This supports the reported failure attribution within these runs without implying that autonomous decisions are universally error-free or that escalation always resolves a task.

C.7 Variation between repeated executions

Table S13 compares the main frontier run, M1, with the E6 Qwen execution on the same 100 tasks. B_0 success changes from 0.88 to 0.91, with six pre-

viously successful tasks lost, and nine previously unsuccessful tasks gained. The total number of changed outcomes is therefore 15, despite a net change of only three tasks. Strong calls per task also change from 4.10 to 3.95.

The configurations differ only in the price table and Jev retry settings, neither of which affects B_0 inference. The B_0 comparison, therefore, shows variation under the same prompts and temperature-zero condition. REFLEX retains 0.95 success in both runs, but four successes are lost and four gained, while its autonomous subset changes from 65 to 66 tasks. Because Jev retry settings differ, these REFLEX changes should not all be attributed solely to provider non-determinism.

Each study is evaluated against its own baseline. The observed three-percentage-point B_0 change is comparable in scale to the E6 success differences and limits the interpretation of small quality changes from single executions. Two runs do not provide a general estimate of run-to-run variance or justify a fixed uncertainty band.

D Extended Cost Results and Claim Register

Table S14 provides the full REFLEX-Sim threshold sweep, including cost per task, cost per suc-

Decision type	Gold kind	Decisions	Errors	Accuracy
risk	control	240	117	0.512
clarification	control	240	75	0.688
completion	terminate	240	16	0.933
retrieval	gather	240	0	1.000
tool selection	gather	240	0	1.000
workflow	commit	240	0	1.000
all		1440	208	0.856

Table S9: **RF-5C errors by decision type.** 192 of 208 errors (92%) fall on decisions whose gold action is a *control* action (risk and clarification); the rest are completion decisions. Retrieval, ordinary tool selection and workflow are at a perfect ceiling across all 720 of their decisions, which is why the effective corpus is 21 families rather than 60. Each of the 12 factorial cells contains 120 decisions (60 families $\times$ 2 surface realizations); deferral accounts for 24 of the 208 errors here (11.5%, a lower bound) against 59.8% of 366 errors in the exploratory study, which is the disagreement that retired C4.

Procedural distance to gold	n	Median sim.	Q1	Q3
$d=1$ (one change from correct)	180	0.310	0.212	0.508
$d=2$ (two changes from correct)	120	0.333	0.206	0.492
$d=\infty$ (far: no edit makes it correct)	5631	0.237	0.166	0.309

Table S10: **Embedding similarity does not order the procedural-distance ladder.** Cosine similarity to the gold action for all 5,931 candidate actions in the 60 RF-5C families, under all-MiniLM-L6-v2. A $d=2$ competitor is *more* gold-similar at the median than a $d=1$ competitor, so an encoder cannot even rank the two ambiguity levels, and 22.2% of far distractors are more gold-similar than the median near competitor (0.322). This is why ambiguity is defined procedurally and not by similarity.

cessful task, and strong calls per successful task. At $\tau = 0.5$, REFLEX costs \$0.00763 per task and \$0.00803 per successful task, compared with \$0.01508 and \$0.01713 for B0. Strong calls per successful task decrease from 4.66 to 1.18. These measures complement call reduction by accounting for task outcomes.

Costs are reconstructed from frozen token counts using the pinned price table. REFLEX-Sim has no external user simulator, so these costs do not include simulator overhead. Per-condition success intervals are absent from the frozen output and are left unspecified. They are not reconstructed from paired-difference intervals, which describe a different quantity.

Table S15 gives the complete claim register. Four of the 12 tested claims were retired, including the proposed interaction, the general deferral tendency, superiority over the cheap self-escalation baseline, and the benefit of hierarchical routing at large candidate-set sizes. A retired claim was not supported by the corresponding evidence and should not be read as proof of its opposite.

The cross-family claim concerns large reductions in calls without a statistically significant decrease in success. The τ^2 -bench claim concerns lower cost, with an unresolved difference in suc-

cess. Both retain the uncertainty described above. The autonomous-subset claim applies to the same 66 tasks under a single frozen gate across the three E6 fallback families, rather than to three independent sets of 66 tasks.

Condition	No failure	Terminal DB	Communication	Control	Arguments
B0 Max-only	54	6	0	0	0
B1 Flash-only	46	14	0	0	0
B3 self-escalation	55	4	1	0	0
R τ $\tau=0.5$	51	8	1	0	0
R τ $\tau=0.8$	49	9	2	0	0
all	255	41	4	0	0

Table S11: **Episode-level failure audit**, all 300 validation episodes across three τ^2 -bench domains. Every failure is either a terminal database-state mismatch or a communication failure. *Not one* episode in any condition failed because a required tool was never called (control) or was called with arguments that failed its action check. A benchmark whose failures are not routing failures cannot demonstrate a routing layer’s value, however cheap that layer is.

Fallback family	95% CI on Δ success	Formal NI at	Escalated-episode steps	Failures	
		−0.02?	still taken by the gate	total	after a gate error
Qwen3.8-Max	[−0.010, +0.090]	yes	26.2%	5	0
Kimi K3	[−0.050, +0.000]	no	26.8%	2	0
DeepSeek-V4-Pro	[+0.000, +0.000]	yes	21.3%	14	0

Table S12: **What E6 does and does not establish**. *Formal NI* asks whether the lower confidence bound clears the predeclared −0.02 margin; the preregistered gate instead compared the point estimate to it, which is why Kimi passes the gate but not a formal test. *Escalated-episode steps still taken by the gate* shows that the compute saving is not only the 66 fully autonomous episodes: even in episodes that escalate at least once, a fifth to a quarter of steps never reach the strong model. *Failures after a bad gate call* counts failures containing at least one incorrect autonomous decision: across all three families it is zero, so every observed failure is attributable to the strong fallback’s own handling after escalation, not to a decision the gate took on its own.

Condition	Success		Tasks that flip		Autonomous
	M1	E6	total	direction	M1 / E6
B_0 Max-only	0.88	0.91	15	6 lost / 9 gained	–
REFLEX $\tau=0.5$	0.95	0.95	8	4 lost / 4 gained	65 / 66

Table S13: **The same condition, executed twice**. The M1 frontier and the E6 Qwen family are separate executions of identical conditions on the identical 100 frozen tasks, at temperature 0, differing only in the price table and in Jev retry settings that B_0 never exercises. They disagree by 3 points of success and 0.15 strong calls per task because the provider is not bit-reproducible. We report each study against its own baseline and never compare a number across the two, because this run-to-run band is as large as the E6 effects themselves.

Condition	Success	95% CI	Strong calls/task	GMR	Cost/task (\$)	Cost/success (\$)	Strong calls/success
B0 Max-only	0.88	–	4.10	–	0.01508	0.01713	4.66
B1 Flash-only	0.87	–	0.00	1.000	0.00120	0.00138	0.00
B3 Flash→Max	0.87	–	2.04	0.502	0.00960	0.01103	2.34
R1 $\tau=0.5$	0.95	–	1.12	0.727	0.00763	0.00803	1.18
R1 $\tau=0.7$	0.91	–	1.30	0.683	0.00868	0.00954	1.43
R1 $\tau=0.8$	0.91	–	1.93	0.529	0.01083	0.01190	2.12
R1 $\tau=0.9$	0.88	–	2.18	0.468	0.01123	0.01276	2.48
R1 $\tau=0.95$	0.88	–	2.30	0.439	0.01158	0.01316	2.61

Table S14: **Extended REFLEX-SIM frontier**. Costs are reconstructed from frozen token counts at the pinned price table. Per-condition success intervals are not available in the frozen output and are not derived from paired-difference intervals, so they are left blank. REFLEX-SIM has no external user simulator, so no simulator cost is folded in.

Claim		
✓	C1	Action-space cardinality degrades a cheap System-One router, with decision ambiguity held fixed.
✓	C2	Decision ambiguity, defined as model-independent procedural counterfactual distance, degrades the same router.
✗	C3	Cardinality and ambiguity are sub-additive (positive interaction).
✗	C4	The characteristic failure of a cheap router is deferral: gathering information instead of acting.
✓	C5	Competitor shape determines WHICH error a router makes without changing HOW OFTEN it errs: swapping one near-miss from a read to a write multiplies unwarranted irreversible commits 6x.
✓	C6	On externally authored function schemas, tool selection is insensitive to how many irrelevant functions are offered, up to K=64.
✓	C7	The hard decision is WHETHER to act, not WHICH tool to call. Same router, same functions: 0.984 choosing a function vs 0.520 deciding whether any applies.
✓	C8	A frozen Jev control layer cuts strong-model calls by 2/3 with no significant success loss, for three unrelated strong-model families.
✓	C9	The confidence gate’s autonomous set is error-free: 66/100 episodes handled without the strong model, zero failures, in all three families.
✓	C10	End to end on τ^2 -bench, REFLEX shows no significant success difference from a strong-only baseline at 3.7x lower cost.
✗	C11	REFLEX beats the cheap self-escalation baseline (B3) end to end on τ^2 -bench.
✗	C12	Hierarchical (two-stage) routing helps at large action-space sizes.

Table S15: **Every claim this work tested, supported (✓) and retired (✗).** Verdicts are recomputed from the frozen manifests by scripts/consolidate_claims.py, which fails if the register and the evidence disagree. 4 of 12 claims were retired, including two the exploratory study appeared to support.